SECTION: DEBUGGING QUESTIONS

For each program, identify the error(s) (syntax and/or logical), explain why it is wrong, and write the corrected code.

D1.

```
#include <stdio.h>
int main() {
    int a, b
    printf("Enter two numbers: ");
    scanf("%d %d", a, b);
    if (a > b)
    printf("A is greater);
    return 0;
}
```

D2.

```
#include <stdio.h>
int main() {
    int x = 10;
    if (x = 5)
        printf("x is 5");
    else
        printf("x is not 5");
    return 0;
}
```

D3.

```
#include <stdio.h>
int main() {
    float pi = 3.14;
    int radius = 5;
    float area;
    area = pi * radius * radius
    printf("Area = %d", area);
    return 0;
}
```

D4.

```
#include <stdio.h>
int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", num);
    if (num % 2 = 0)
        printf("Even");
    else
        printf("Odd");
```

```
    return 0;
}
```

D5.

```
#include <stdio.h>
int main() {
    int marks;
    scanf("%d", &marks);

    if (marks >= 90);
        printf("Grade A");
    else if (marks >= 75)
        printf("Grade B");
    else
        printf("Grade C");

    return 0;
}
```

(Hint: look carefully at the semicolon right after the `if` condition.)

D6.

```
#include <stdio.h>
int main() {
    char grade = 'A'
    int marks = 85;

    if (marks > 90)
    {
        grade = 'A';
    else if (marks > 75) {
        grade = 'B';
    }

    printf("Grade: %c", grade);
    return 0;
}
```

D7. This program is supposed to swap two numbers using a temporary variable but gives the wrong output. Find the bug:

```
#include <stdio.h>
int main() {
    int a = 5, b = 10, temp;
    temp = a;
    a = b;
    b = temp;
    printf("a = %d, b = %d", a, a);
    return 0;
}
```

D8. The program should print whether a number is positive, negative, or zero, but always prints "Positive" for negative numbers too. Find and fix the logic error:

```
#include <stdio.h>
int main() {
    int n = -5;
    if (n >= -100)
        printf("Positive");
    else if (n < 0)
        printf("Negative");
    else
        printf("Zero");
    return 0;
}
```

D9. This program is meant to print only the name of the day matching `day`, but for `day = 3` it prints `"WednesdayThursdayInvalid"` all together. Find the bug and fix it:

```
#include <stdio.h>
int main() {
    int day = 3;
    switch (day) {
        case 1:
            printf("Monday");
        case 2:
            printf("Tuesday");
        case 3:
            printf("Wednesday");
        case 4:
            printf("Thursday");
        default:
            printf("Invalid");
    }
    return 0;
}
```

D10. A student tries to use `switch` to compare a floating-point value and is confused why it won't even compile. Identify the exact rule being violated, and rewrite the logic correctly using `if-else` instead (do not simply try to "force" `switch` to work):

```
#include <stdio.h>
int main() {
    float x = 2.5;
    switch (x) {
        case 2.5:
            printf("Two point five");
            break;
        default:
            printf("Other");
    }
    return 0;
}
```

D11. For `a = 10, b = 20, c = 5`, this program is supposed to print `"A is largest"` only when `a` is greater than both `b` and `c`, and print `"something else"` otherwise. It compiles fine but gives a logically wrong/misleading result. Trace through it, identify which `if` the `else` actually binds to, and fix it using proper braces:

```
#include <stdio.h>
int main() {
    int a = 10, b = 20, c = 5;
    if (a > b)
        if (a > c)
            printf("A is largest");
    else
        printf("something else");
    return 0;
}
```